# Gagan Jain

**AI / LLM Engineer**

Jaipur, India · +91 95872 55792 · gagan.jain.se@gmail.com

## PROFESSIONAL SUMMARY

AI Engineer focused on production-grade GenAI systems: agentic AI orchestration, RAG pipelines, LLM fine-tuning (QLoRA/LoRA), MCP tooling, and local-first LLM deployment. Author of NexusAOS, a governance-first, event-sourced AI operating environment written in Rust (12 workspace crates, 981 tests, full CI/CD) with provider-agnostic LLM streaming (OpenAI/Anthropic), policy-enforced agent actions, and terminal/SSH integration. Strong Python + Rust foundation with end-to-end deployment: Docker, GitHub Actions, observability, and AI governance.

## TECHNICAL SKILLS

### AI / LLM

LLM Integration, RAG Pipelines, Agentic AI Frameworks, Prompt Engineering, Context Engineering, Multi-Agent Systems, Tool-Using Agents, Fine-tuning (QLoRA/LoRA), MCP / Tool Protocols, LangChain / LlamaIndex, Vector Databases, AI Governance

### LLM OPS

Evaluation & Observability, Local LLM Deployment, OpenAI / Anthropic APIs, Embeddings, Structured Outputs / Function Calling, LiteLLM / Model Routing, Streaming Responses, NLP, Ollama

### LANGUAGES

Python, TypeScript, JavaScript, Rust, C++, C, PHP, HTML/CSS

### FRONTEND

React, React Native, Tailwind CSS, Bootstrap, Monaco Editor, Next.js

### BACKEND

Node.js, Flask, Express, FastAPI, Axum (Rust), REST APIs, WebSockets, Laravel, Supabase

### DATABASES

MySQL, MongoDB, Redis, SQLite

### DEVOPS

Docker, GitHub Actions, Nginx, Gunicorn, Prometheus, Linux

### VISION / ML

YOLO, OpenCV, Computer Vision

## PROJECTS

### NexusAOS AI/AGENTIC

Governance-first, event-sourced AI operating environment in Rust — 12 workspace crates, 981 tests, provider-agnostic LLM streaming.

- Designed a governance-first, event-sourced AI OS in Rust: 12 workspace crates, 981 passing tests, 0 clippy warnings, full GitHub Actions CI/CD
- Built a policy-enforced agent kernel where LLMs propose actions and the kernel validates and records every state change in an append-only audit trail
- Implemented a provider-agnostic LLM layer with OpenAI-compatible and Anthropic streaming, LiteLLM routing, and local-first inference that runs fully offline
- Delivered native terminal emulation (PTY + VT100), SSH multiplexing, a secrets vault, and four interfaces: CLI, TUI, GUI, RPC

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### SeshaOS AI/AGENTIC

NexusAOS v2 — governance-first, local-first AI OS with specialist local models and LiteLLM-compatible routing.

- Architected specialist model stack: Gemma 4 12B (Planner), Qwen3-Coder 30B (Implementation), Qwen3.5 9B (Vision)
- Designed kernel-centric governance where models propose actions and the kernel validates/records every state change
- Built a LiteLLM-compatible proxy for NVIDIA NIM and other model providers
- Maintained event-sourced architecture with reversible, permissioned actions and offline-first execution

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### Vyākṛti FLAGSHIP

Sanskrit-oriented programming language with complete compiler pipeline and browser-based IDE. 123 tests.

- Complete compiler pipeline: lexer → parser → type checker → bytecode compiler — all built from scratch in Rust
- Browser-based IDE with React, Monaco Editor, syntax highlighting, autocomplete, and diagnostics
- Rust (axum) backend with compile, REPL, LSP, and file management endpoints via REST + WebSocket
- 123 tests covering the full pipeline, including a self-hosting corpus

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### AIM — Attendance Information Manager PRODUCTION-READY

Production-grade Flask + MySQL platform with Argon2id auth, Prometheus monitoring, and CI/CD.

- Production-grade Flask + MySQL platform with Argon2id auth, CSRF protection, brute-force lockout, breached-password scanning, JWT sessions, and strict CSP/HSTS headers
- Deployed Prometheus metrics, structured JSON logging, Chart.js analytics, FullCalendar scheduling, Docker Compose, and GitHub Actions CI/CD; 101 automated pytest tests

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### RAG Service — Production RAG + Vector Search GEN AI

Production RAG API service: hybrid retrieval (vector + keyword), embeddings, reranking-ready, and an LLM-as-judge eval harness.

- Built a FastAPI RAG service with hybrid retrieval (dense embeddings + BM25 keyword, merged via Reciprocal Rank Fusion) over ChromaDB
- Implemented chunking strategies (recursive, semantic), embedding pipelines, and an /ask endpoint that streams grounded answers with citations
- Shipped LLM-as-judge evaluation harness (faithfulness, answer relevance, correctness) with golden-set CI gate and RAGAS-style reports
- Containerized with Docker Compose; OpenAI-compatible LLM interface with local-first fallback

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### LLM Eval Harness GEN AI

Reusable LLM evaluation harness: golden-set YAML, LLM-as-judge metrics, offline heuristic fallbacks, and CI-ready reports.

- Built a golden-set-driven evaluation harness with faithfulness, answer-relevance, and correctness metrics (LLM-as-judge + lexical fallbacks)
- Added offline heuristic scorers so evals run in CI without API keys; JSON + Markdown reports for regression tracking
- Designed golden-set YAML format (question, golden answer, context) and a CLI: python -m eval_harness --golden-set ...
- Containerized and CI-ready; used to gate the RAG service pipeline

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## EXPERIENCE

### Industry 5.0 Industrial Automation Trainee

Aug – Nov 2025

**CodenPlay Robotics**

VIT Vellore / Online

- Built and debugged PLC automation logic (ladder logic, structured text) in CODESYS
- Simulated industrial process workflows in Factory I/O
- Implemented flow-based control/integration in Node-RED
- Covered HMI design, industrial communication, process monitoring, and the debugging→testing→deployment workflow

## EDUCATION

### VIT Vellore

Graduated 2025

B.Tech in Computer Science and Engineering